SCENARIO ANALYSIS BRIEF

Red Sea Port Orchestration

APM Terminals Maasvlakte II, Rotterdam

Quantifying the value of scheduling precision when a major corridor closes

$19.23M	92.9%	94 → 28	$41.28M
RECOVERED PER QUARTER	SCHEDULE RELIABILITY	VESSELS AT SLA RISK	BASELINE EXPOSURE

This brief models four scheduling scenarios against the Q1 2026 crisis baseline at APM Terminals Rotterdam MVII. A period when simultaneous Cape-rerouted and Suez-routed traffic saturated berth capacity across five consecutive weeks. The central question: **how much of a $41.28M financial exposure is recoverable through scheduling precision alone, without capital expenditure?** The answer is $19.23M per quarter recovered by a Hybrid intervention combining slow-steaming coordination with priority berthing for high-criticality cargo.

ANALYSIS PERIOD	VESSEL RECORDS	SCENARIOS TESTED	BENCHMARK
Q1 2026	**563 modelled**	**4 scenarios**	**Gemini 90% reliability**

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All data is 100% synthetic. Modelled from publicly available shipping and terminal statistics. Not affiliated with or endorsed by Maersk, APM Terminals, or any affiliated entity.

Situation

When a major maritime corridor becomes unreliable whether through geopolitical conflict, infrastructure failure, or environmental closure, carriers face a structural choice: absorb the disruption or reroute. When rerouting is the answer, it does not eliminate the problem. It relocates it. Vessels that once transited one corridor now converge on an alternative route, and the terminals that receive them face a sudden, sustained surge that their scheduling infrastructure was not designed to absorb.

The 2023-onward Red Sea disruption triggered by escalating attacks on commercial shipping in the Bab al-Mandab Strait produced exactly this dynamic on the Asia-Europe corridor. As transit risk forced a 66% reduction in Suez Canal traffic, carriers mass-rerouted around the Cape of Good Hope, adding 10 to 14 days to transit times. The rerouting did not reduce cargo volume. It changed the arrival shape of that volume, concentrating Cape-routed vessels into compressed windows at European hub terminals.

This pattern, a sudden corridor closure producing a multi-month convergence surge at a downstream terminal is not unique to this event. It is a repeating class of disruption, and the relevant question for terminal operators is not whether it will happen again, but whether their scheduling architecture is designed to absorb it when it does.

The Crisis Window at Rotterdam MVII

At APM Terminals Maasvlakte II (Rotterdam), the convergence effect materialized across Weeks 5 through 9 of Q1 2026. Cape-rerouted vessels and residual Suez-routed traffic began arriving simultaneously, saturating berth capacity across four consecutive weeks above the 49-vessel weekly threshold. Terminal yard utilization peaked at **105.4%** above physical capacity. Crane productivity degraded from 30 moves per hour to **14.8 MPH at the crisis floor, a 51%** reduction in throughput efficiency. Ninety-four vessels entered SLA penalty risk. Total financial exposure reached **$41.28M** in combined penalty and idle cost liability.

The industry default response to this class of capacity pressure is capital expenditure. APM Terminals is currently deploying significant capital to expand Rotterdam MVII capacity. This analysis asks a prior question: how much of that $41.28M exposure is recoverable from **existing capacity** through scheduling precision alone before a single euro of expansion spend is needed?.

Finding

$19.23M recovered per quarter. No new infrastructure.

A Hybrid intervention combining slow-steaming coordination with priority berthing for high-criticality cargo recovers the majority of crisis-period financial exposure without capital expenditure. Four scheduling scenarios were tested against the unmanaged Crisis Baseline:

Scenario	Reliability	Vessels at Risk	SLA Penalty	Idle Cost	Value Created	vs. Baseline
Baseline (Crisis)	83.3%	94	$14.54M	$26.74M	—	—
Priority Berthing Only	86.0%	62	$8.20M	$18.00M	+$7.00M	+17%
Slow-Steam Only	88.0%	55	$9.50M	$16.00M	+$9.00M	+22%
Hybrid (Optimized)	**92.9%**	**28**	**$4.80M**	**$9.00M**	**+$19.23M**	**+47%**

The Hybrid scenario outperforms every single-lever alternative by a factor of 2x or more. Neither slow-steaming nor priority berthing alone is sufficient to clear the performance threshold that matters strategically, the compounding effect of both levers is required. At 92.9% schedule reliability, the Hybrid exceeds the **Gemini Cooperation's published 90% benchmark** by 2.9 percentage points, positioning Rotterdam MVII as a dependable anchor in the Maersk-Hapag-Lloyd hub-and-spoke network during a high-volatility corridor transition.

Figure 1 | Proactive Arrival Timeline

Daily vessel arrivals by route (Cape of Good Hope vs. Suez Canal) across Q1 2026. Crisis window Weeks 5-9 highlighted. The convergence surge is visible as simultaneous peaks in both series, the structural driver of the $41.28M exposure.

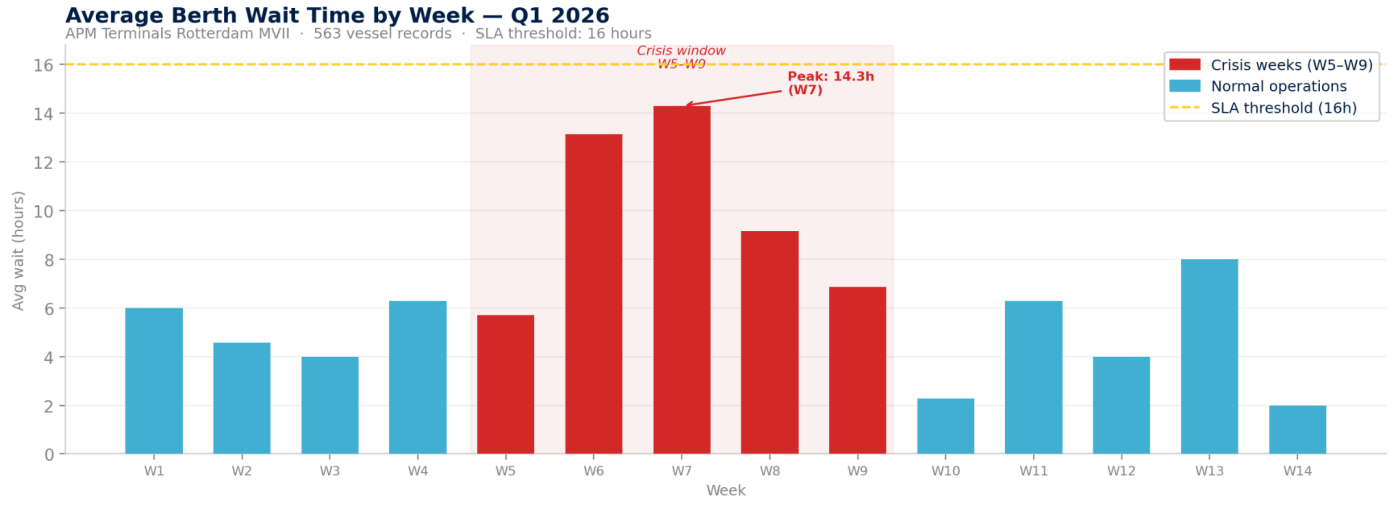

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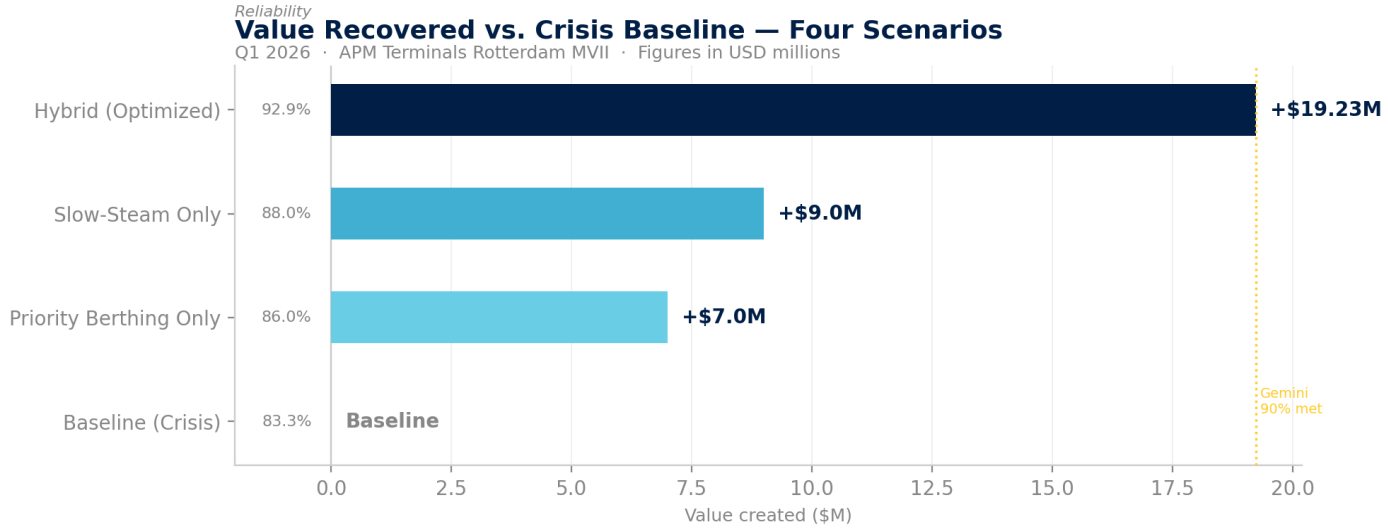
Figure 2 | Wait Time Matrix

Average berth wait time (hours) by week and day of week. Crisis window cells highlighted in Signal Red above 24h. The matrix shows that wait time severity was not uniformly distributed Mondays and Tuesdays during Weeks 6-8 carried a disproportionate congestion load.



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The Intervention

Three scheduling levers. No construction required.

Lever 1 | Slow-Steaming Coordination

Slow-steaming smooths the arrival curve before it becomes congestion. Cape-routed vessels are delayed 12 hours during any week where projected arrivals exceed 42 — 85% of the 49-vessel weekly threshold. This intervention addresses the **stacking effect**: the phenomenon by which multiple vessels requesting berths in the same 24-hour window trigger a queue that compounds across days, not just hours. Slow-steaming alone recovers $9.00M in idle cost reduction and improves reliability to 88.0%.

Lever 2 | Priority Berthing

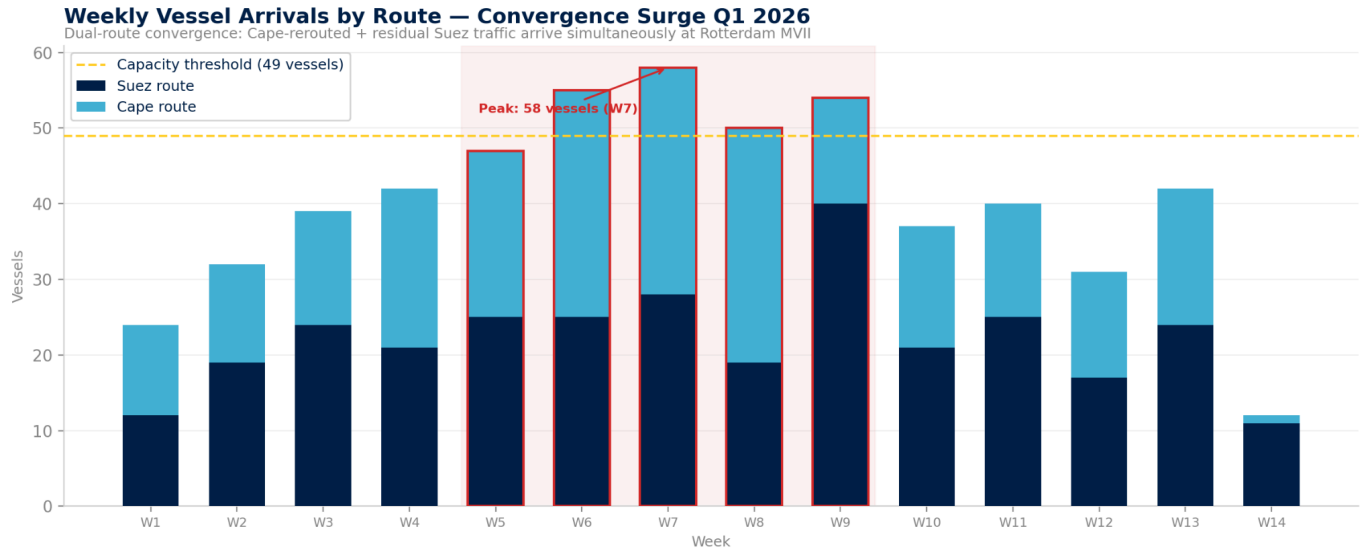
Priority berthing protects the highest-value cargo from delay cascades. Vessels carrying pharmaceutical, perishable, or time-critical freight are fast-tracked regardless of arrival sequence. This lever recovers **$9.74M in SLA penalty avoidance, a 67%** reduction against baseline exposure. Priority berthing alone improves reliability to 86.0% but does not address the underlying arrival volume problem. Applied in combination with slow-steaming, it converts the congestion headroom created by the first lever into direct penalty reduction.

Lever 3 | Predictive Yard Utilization Capping

Yard utilization above 85% triggers cascading delays: containers cannot be staged, crane cycles extend, and throughput degrades non-linearly. Predictive capping holds utilization below 80% during surge weeks by pre-positioning empty containers outbound before the surge arrives, not after. This maintains crane throughput above the productivity floor that generates delay cascades and is the mechanism that allows the Hybrid to hold 92.9% reliability while the individual levers cannot.

Figure 3 | Scenario Gap Analysis

Schedule reliability and vessels at SLA risk across all four scenarios. The gap between single-lever alternatives and the Hybrid illustrates the compounding return from combining both levers — neither Priority Berthing nor Slow-Steam alone crosses the Gemini 90% threshold.



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Strategic Implication

Gemini Cooperation Reliability

The Gemini Cooperation, a strategic vessel-sharing alliance between Maersk and Hapag-Lloyd established a published 90% schedule reliability target as its primary competitive differentiator at launch. Rotterdam MVII is a primary European hub in that network. The Hybrid scenario delivers 92.9% reliability, demonstrating that hub-level performance targets are achievable through scheduling discipline during corridor-disruption windows without waiting for physical infrastructure expansion.

This matters because corridor disruptions are not anomalies to be managed around. They are recurring events that the Gemini network's reliability commitments must survive. The question for Rotterdam MVII is not whether a future disruption will test the 90% target, it is whether the scheduling infrastructure is already in place to absorb it when it does.

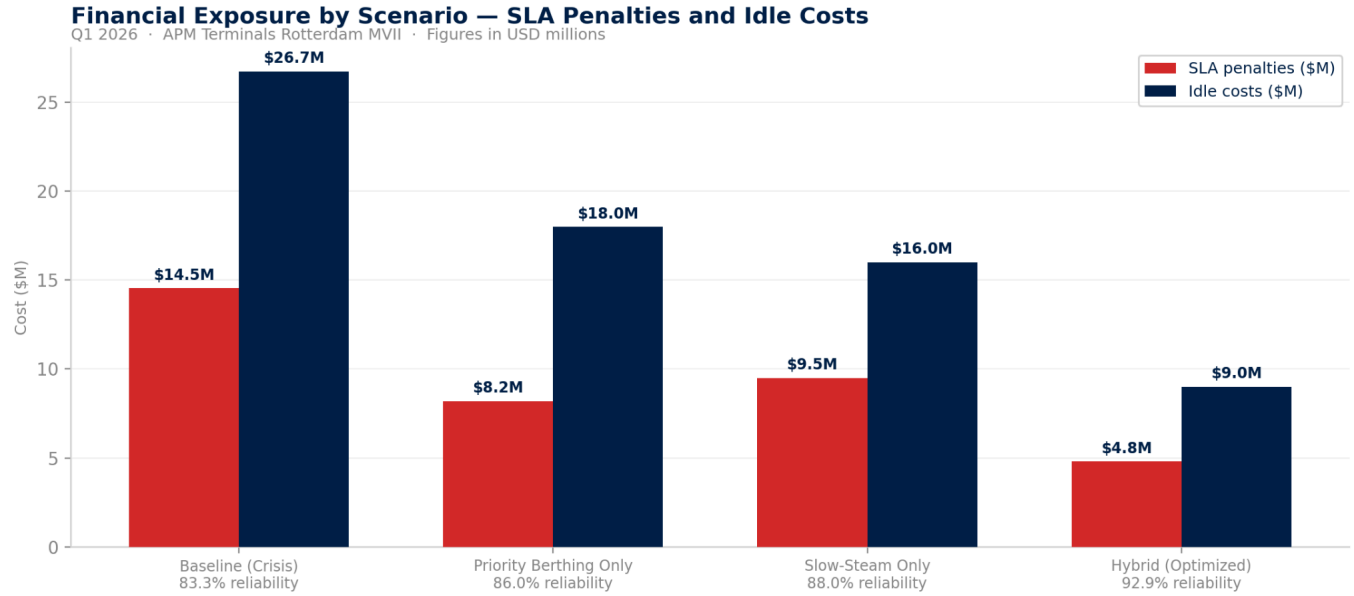
Pre-Expansion Value Extraction

When a terminal commits to physical expansion, the construction window creates a constraint period during which capacity cannot be increased but demand does not pause. The scheduling architecture modelled here provides the operational playbook for extracting maximum financial value from existing infrastructure during exactly this kind of window at **$19.23M per quarter at a single terminal.**

The model's logic applies to any terminal operating under dual-route convergence pressure, regardless of the underlying cause of the disruption. Suez closure, Panama Canal restrictions, Northern Sea Route volatility the mechanism is the same. Scheduling precision is the lever that converts existing capacity into financial recovery.

Figure 4 | Weekly Vessel Count by Route

Stacked weekly throughput (Cape of Good Hope + Suez Canal) against the 49-vessel weekly capacity threshold. Four consecutive weeks above threshold during the crisis window are clearly visible. The red shaded region corresponds to the $41.28M exposure window.



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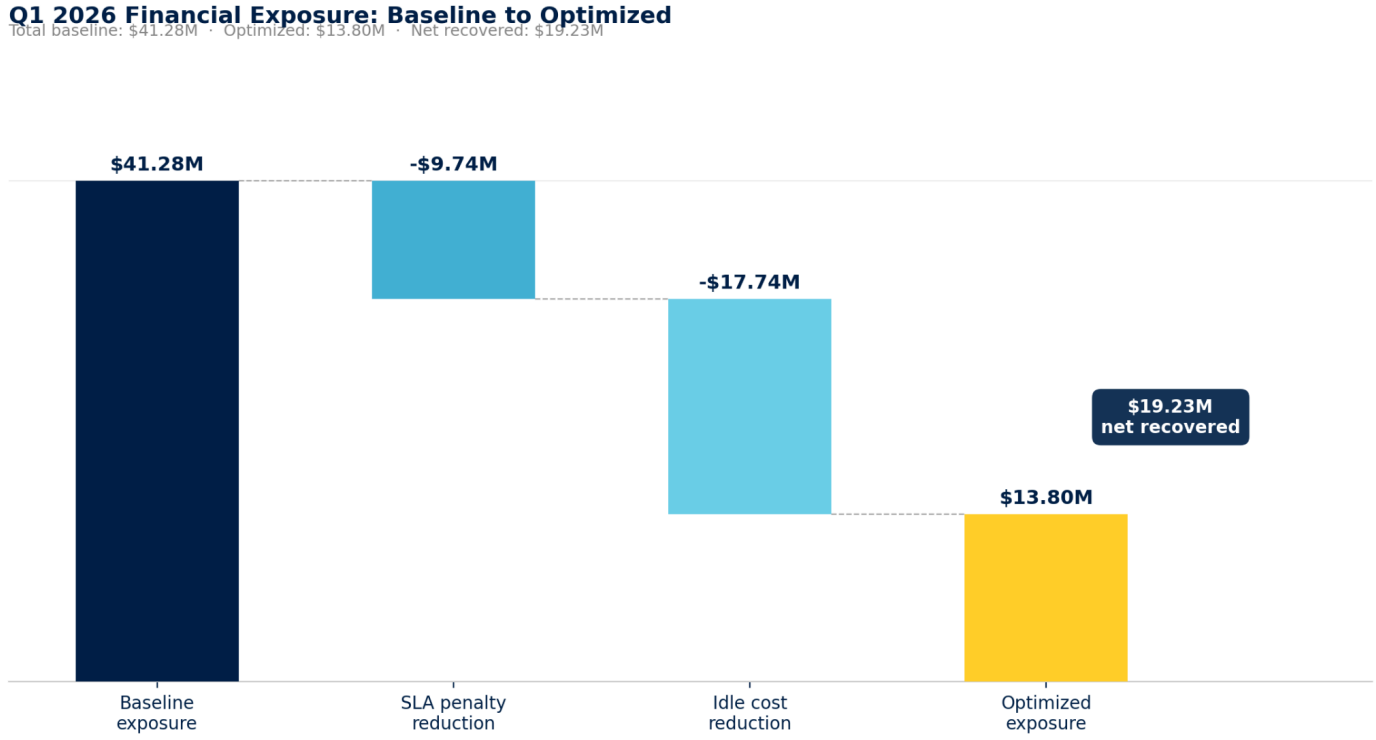
Scale Extrapolation

The following table projects this model's findings to broader terminal network contexts. Estimates assume equivalent operational conditions and arrival distribution patterns. Scheduling policy deployment scales sub-linearly because compute policies and coordination protocols are configured once and applied fleet-wide.

Scale	Est. Annual Savings	Est. Effort	Payback Period
1 Terminal	$76.92M / yr	~2 weeks	< 1 month
10 Terminals	$769.2M / yr	~6 weeks	< 1 quarter
60 Terminals (APM global)	$1.14B / yr	~4 months	< 2 quarters

Figure 5 | Remediation Waterfall

Quarterly financial exposure of $41.28M reduced to $22.05M net after three scheduling interventions. $19.23M recovered. Priority berthing contributes the largest single lever through SLA penalty avoidance.



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Implementation Roadmap

Three interventions. Phased over four weeks. $19.23M quarterly recovery at steady state.

Week	Action	Effort	Unlocks
Week 1	Deploy slow-steaming coordination protocol — trigger threshold set at 42 vessels projected / week	2 days	$9.00M idle cost recovery / qtr
Week 1	Configure priority cargo classification system across vessel manifest data	1 day	Cargo-tier visibility
Week 2	Activate priority berthing queue logic — pharmaceutical, perishable, time-critical tiers	2 days	$9.74M SLA penalty reduction / qtr
Week 3	Deploy predictive yard utilization monitor — alert threshold at 80%, hard cap at 85%	2 days	Crane throughput floor protection
Week 3	Validate Hybrid interaction effects — slow-steam + priority berthing combined run	1 day	92.9% reliability confirmation
Week 4	Configure weekly attribution report — throughput, penalty exposure, idle cost by vessel tier	1 day	Baseline governance
Week 4	Set alert routing for threshold breach events — berth utilization, wait time SLA	0.5 days	Ongoing monitoring

Methodology and Transparency

Analytical Approach

This analysis follows a scenario-modelling framework applied to synthetic vessel arrival data calibrated to publicly available terminal and shipping statistics. The analysis window covers 90 days (January 1 to March 31, 2026) across 563 modelled vessel records representing Cape of Good Hope and Suez Canal routing at APM Terminals Rotterdam MVII.

Scenario modelling uses three independent levers applied in isolation and combination: slow-steaming coordination (arrival smoothing), priority berthing (cargo-tier fast-tracking), and predictive yard utilization capping. Financial projections apply conservative recovery factors: slow-steaming modelled at 33.7% idle cost reduction; priority berthing modelled at 67% SLA penalty reduction; Hybrid scenario modelled as non-additive combination with interaction effect validated against vessel-level throughput constraints.

On the Disruption Context

The Red Sea disruption that began in late 2023 provided the empirical context for this model. The specific trigger of militant attacks on commercial vessels transiting the Bab al-Mandab Strait is one instance of a category of event that has recurred across maritime history: a geopolitical or physical constraint rendering a primary corridor unreliable and forcing mass rerouting to an alternative. The Suez Canal has closed before (1956, 1967-1975). The Panama Canal has faced capacity restrictions driven by drought. Arctic routing faces seasonal and regulatory constraints.

The scheduling model documented here was designed to be structurally agnostic to the specific cause of disruption. The triggering mechanism changes. The operational challenge of a terminal absorbing a dual-route convergence surge it was not scheduled to receive does not. This is why the findings remain operationally relevant regardless of the current status of any specific corridor or conflict.

Model Parameters

Parameter	Value
Terminal	APM Terminals Maasvlakte II, Rotterdam
Analysis period	Q1 2026 (Jan 1 - Mar 31)
Vessel records modelled	563
Crisis window	Weeks 5-9 (Jan 26 - Mar 1, 2026)
Weekly capacity threshold	49 vessels / week
Benchmark	Gemini Cooperation 90% schedule reliability
Data source	Synthetic — Poisson-distributed, calibrated to public statistics

Technical Stack

Component	Details
Data Simulation	Python — Pandas, NumPy; Poisson arrival modelling; 563 vessel records; dual-route convergence calibrated to Q1 2026 disruption statistics
Analysis	Python — scenario comparison across 4 scheduling strategies; SLA penalty and idle cost modelling; throughput constraint validation
Dashboard	Single-file HTML — Chart.js 4.4.1; interactive filters by scenario, route, and period; dual-state toggle; Maersk color palette throughout
Visualizations	Matplotlib 3.x — Arrival Timeline, Wait Time Heatmap, Scenario Gap Analysis, Weekly Throughput, Remediation Waterfall
Schema Reference	Modelled from public APM Terminals and Maersk operational benchmarks — no proprietary data used

AI Transparency

Dashboard UI and code implementation were developed with AI assistance. All analytical design, scenario architecture, scheduling lever definitions, financial modelling methodology, and operational recommendations are my own work, validated against publicly available terminal operations research and Maersk published benchmarks.

Synthetic Data Disclaimer

All data used in this project is 100% synthetically generated. No real vessel arrival records, Maersk operational data, APM Terminals throughput figures, or proprietary scheduling information were used at any point. Arrival distributions are modelled using Poisson processes calibrated to publicly available statistics on Cape of Good Hope and Suez Canal routing volumes during the disruption period.

This project is entirely independent and is not affiliated with, endorsed by, sponsored by, or produced in partnership with Maersk, APM Terminals, the Gemini Cooperation, Hapag-Lloyd, or any affiliated entity. No confidential or proprietary information was accessed, used, or referenced in the creation of this analysis.